1.Given the binary data sequence 101101101 and the divisor 1101, perform modulo-2 division to find the remainder (CRC code) and write the transm itted codeword.

2. Given the binary words 11010110, 10011001, and 11100010, calculate the 8-bit checksum to be appended to the data for transmission. Show all carries and the final result.

3. **Given the data bits 1011000, generate the even parity (VRC) bit and write the transmitted codeword.**

4. **Given the data matrix:**

text

1100111

1011101

0110101

1010101

Construct the LRC block and write the transmitted row.

5 Given the 7-bit data 1011001, construct the Hamming (11,7) codeword by determining and inserting the correct parity bits (r1, r2, r4, r8). Show the calculation for each parity bit as per the sets in your reference.

6. A UDP header is given as: 85 43 00 35 00 28 1C 3F.

- Identify and convert the Source Port, Destination Port, Length, and Checksum fields from hexadecimal to decimal.

- State the purpose of the checksum in the UDP header.

7.

Given a bit stream 0 1 0 0 1 1, draw the Differential Manchester encoding waveform. Explain how each binary value is represented using transitions and no transitions**.**

8.

If you divide the 192.168.10.0/24 network into 3 equal subnets, what subnet mask must you use? How many subnets and hosts per subnet will result from this division? Show your calculation steps.

2.Given data sequence 111010 and divisor polynomial $x^3 + x^2 + 1$ (which is 1101 in binary), calculate the CRC code appended before transmission.

3.For a data sequence 1000110, use divisor 10011 (corresponding to $x^4 + x + 1$). Do the CRC division and specify the remainder.

4.Given data 110110110 and generator polynomial $x^4 + x^3 + 1$ (binary 11001), find the CRC remainder using modulo-2 division.

5.A message sequence 1011010 is divided by the generator polynomial $x^5 + x^2 + x + 1$ (binary 100111), find the quotient and remainder using CRC division, and write out the codeword for transmission.

6 **Given the binary words 11010110, 10011001, and 11100010, calculate the 8-bit checksum to be appended to the data for transmission. Show all carries and the final result.**

7.For the data segments 10101100, 01110011, and 01010101, perform binary addition, showing any carries, and determine the checksum for error detection purposes.

8.Four binary numbers are received: 01101001, 11000101, 00011010, and 10101010. Add them with carries and find the checksum value to detect errors.

9.Given five 8-bit blocks: 10011001, 11100010, 00100100, 10000100, and 11011010, compute the sender-side checksum. Show the addition and wrap-around/carry handling as in examples.

10. The receiver receives the data blocks: 11010110, 10011001, 11100010, 00100100, and the checksum block 01011001. Add all blocks, including the checksum, and verify if the transmission is error-free by checking if the result is all 1s.

11.**Given the data bits 1011000, generate the even parity (VRC) bit and write the transmitted codeword.**

12. For the data 1101110, determine the odd parity bit and show the full code to be sent.

13. The receiver receives codeword 01011001. Check if the even parity bit is correct, and determine if an error is detected.

14.Given codeword 01101010 (with parity bit), identify whether even or odd parity was used and verify correctness.

15.For data sequence 1010001, calculate both the even and odd parity bits that could be appended for transmission.

16.**Given the data matrix:**

text

1100111

1011101

0110101

1010101

Construct the LRC block and write the transmitted row.

17.Data blocks: 10011001, 01110001, 11100111, 10111011. Find the LRC bits for each column and provide the LRC word.

18 For the following received blocks and LRC:

text

11010101

10011011

00011011

11010010

LRC: 10001010

Verify whether an error is present based on the LRC check.

19 Data blocks: 10101110, 11010101, 01111011, 10001010. Compute the LRC and list parity for each column.

20.If LRC for the following blocks (each 8 bits) is required:

text

10101001

01101101

11100011

11001101

Find the LRC byte to be sent with the data.

1.

Given the 7-bit data 1011001, construct the Hamming (11,7) codeword by determining and inserting the correct parity bits (r1, r2, r4, r8). Show the calculation for each parity bit as per the sets in your reference.

2.

A receiver gets the 11-bit Hamming codeword 10111010010. Using the parity positions r1, r2, r4, r8, check for errors and identify the location of a single-bit error if present.

3.

For the message bits 1100110, compute and write down the corresponding Hamming code including all parity bits. Detail which bits each r bit covers and how each is calculated.

4.

Decode the Hamming codeword 10010011101. Show the parity bit checks, locate any error, correct the erroneous bit, and write the corrected code.

5.Explain how each parity bit position (r1, r2, r4, r8) is calculated for the message bits 1110011 and write the full encoded Hamming codeword for transmission.

1.**A UDP header is given as: 85 43 00 35 00 28 1C 3F.**

- **Identify and convert the Source Port, Destination Port, Length, and Checksum fields from hexadecimal to decimal.**
- **State the purpose of the checksum in the UDP header.**

2.

Given the TCP header section:
00 50 23 4A 5A 2C 1F 00 70 02 40 00 F2 42 00 00

- Identify the Source Port, Destination Port, Sequence Number, and Acknowledgment Number fields.
- Convert all four identified fields from hex to decimal.

3.

Suppose a UDP datagram has the header in hex: 13 57 16 2C 03 40 B8 2E.

- Calculate the decimal values for the Source Port, Destination Port, Length, and Checksum.
- Explain how the Length field is interpreted in UDP communication.

4.

Given the following TCP flags byte in a header: 70 02.

- Interpret the flags (e.g. SYN, ACK, FIN) present.
- State which ports are used for the Source Port and Destination Port if the previous bytes are 00 50 and 23 4A.

5.

A UDP packet received contains the header: A2 4F B3 15 00 64 5F 1B.

- Decode and write the decimal Source Port, Destination Port, Length, and Checksum.
- What is the maximum possible value for the Length field in UDP, and why?

1.

Given a bit stream 0 1 0 0 1 1, draw the Differential Manchester encoding waveform. Explain how each binary value is represented using transitions and no transitions.

2.

For the binary data 1 0 1 1 0 0, illustrate the corresponding Manchester encoding waveform. Describe how the position and direction of transitions identify zeros and ones.

3.

Explain how NRZ-I encoding represents the sequence 0 1 1 0 1 0. Draw its waveform and specify which bits cause level changes.

4. For the sequence 1 1 0 1 0 0, describe the differences in their waveforms and how each encoding technique handles synchronization and bit representation. In both Differential Manchester and Manchester Encoding

1.

If you divide the 192.168.10.0/24 network into 3 equal subnets, what subnet mask must you use? How many subnets and hosts per subnet will result from this division? Show your calculation steps.

2.

List the subnet ID (network address), broadcast address, and range of usable host addresses for the first subnet created when dividing 192.168.10.0/24 into 3 subnets.

3. What are the subnet ID, broadcast address, and usable host IP range for the second subnet when dividing 192.168.10.0/24 into three networks? Demonstrate your calculation.

4. Find the subnet ID, broadcast address, and range of usable host addresses for the third subnet resulting from subnetting 192.168.10.0/24 as described. Explain each step clearly.

5.

If you needed exactly 25 hosts per subnet and the maximum number of subnets from 192.168.10.0/24, what subnet mask would you select? List the network address and broadcast address for each resulting subnet